SIMATS ENGINEERING

ASSESSMENT TOOL 2

CONCEPT MAPPING

Course Name: Operating System
Course Code: CSA0417

Student Name: A. Lakshmi Devi
Register Number: 192525176

Course Outcome Covered

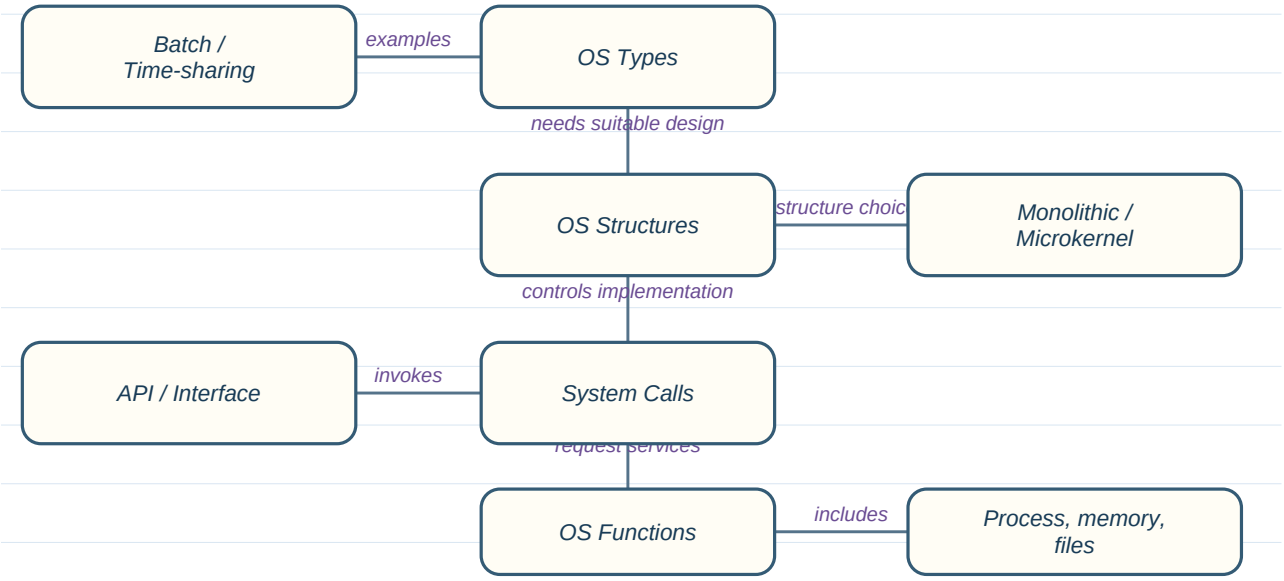
CO1: Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments.

Assessment – CO1 – AT 2
Total Marks: 100

Student Signature: _____

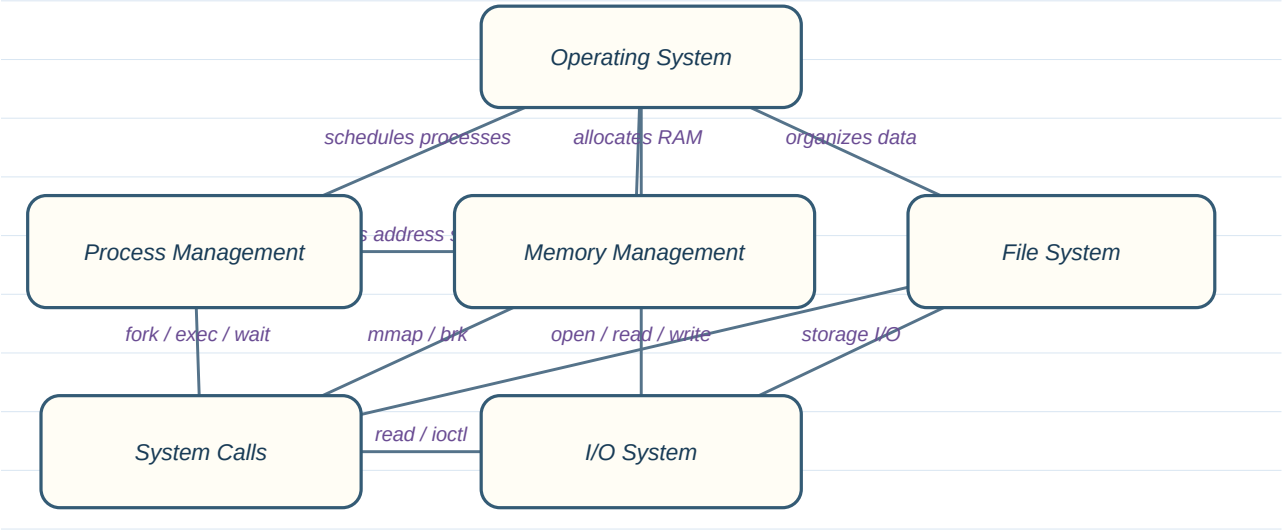
ANNEXURE A — CONCEPT MAPPING

Question 1. Multi-level concept map: OS Types → Structures → System Calls → OS Functions



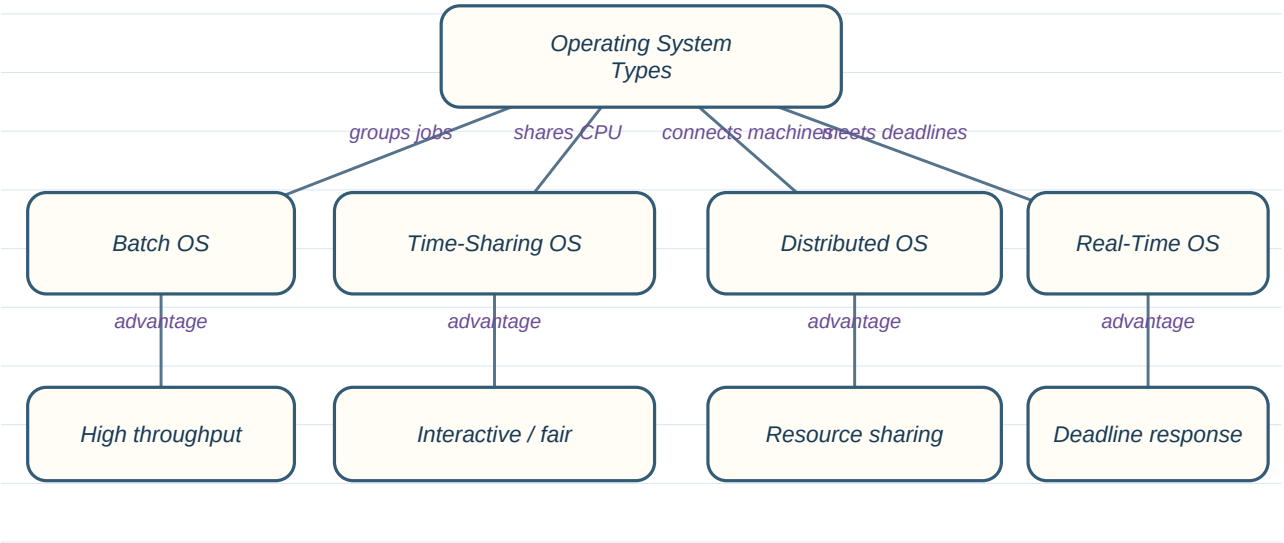
Explanation: *Operating-system types are designed for different workloads. The OS structure determines where services run and how system calls reach the kernel. System calls provide a controlled interface for services such as process creation, memory use, file access and device I/O.*

Question 2. Concept map of major OS management components



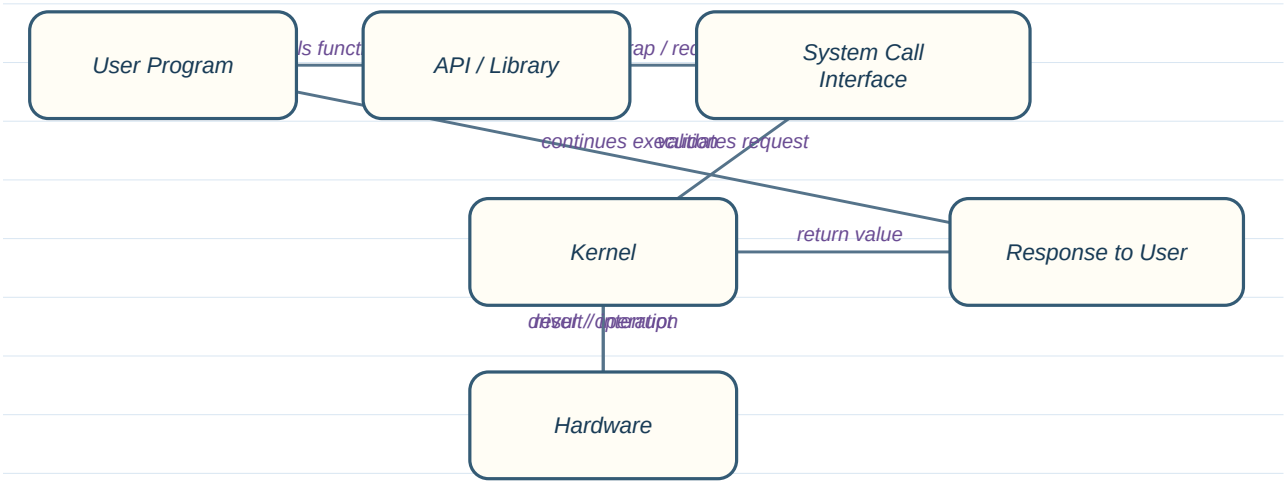
Interdependencies: *A process needs memory to execute. File operations may trigger disk I/O, while device drivers transfer data between hardware and memory. System calls let user programs request these services without directly controlling hardware.*

Question 3. Comparative concept map of OS types



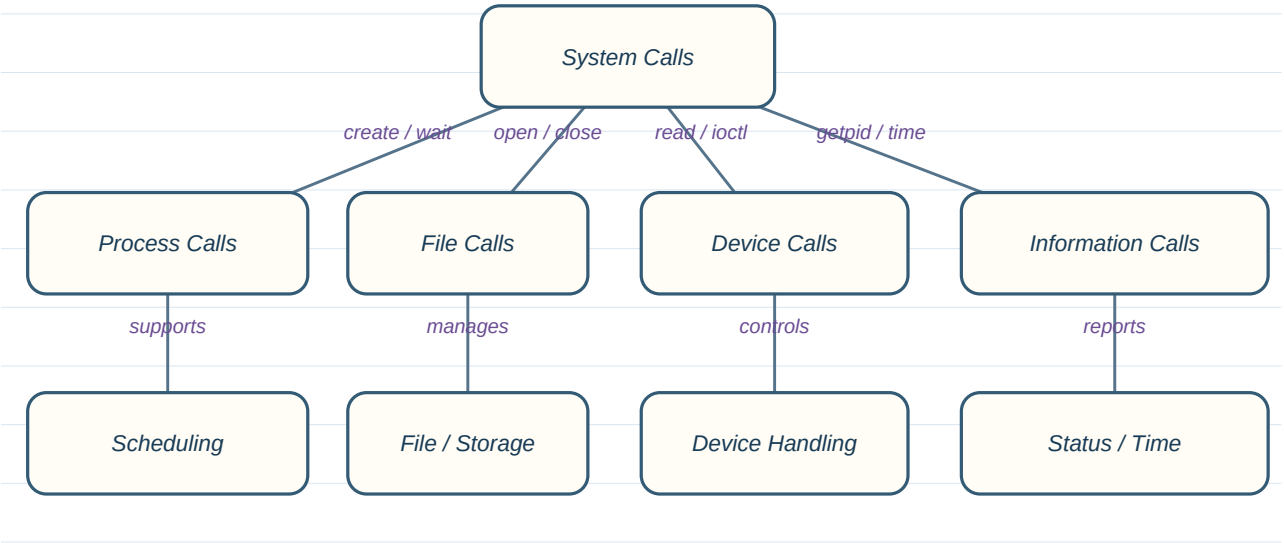
Comparison: *Batch OS executes grouped jobs with little interaction; it is efficient for repetitive work but has slower response. Time-sharing gives many users interactive CPU time, but performance can fall under heavy load. Distributed OS coordinates networked computers and shares resources, but depends on network reliability. Real-time OS prioritizes predictable deadlines; it is useful for control systems, though design and scheduling are strict.*

Question 4. Execution flow of a system call



Control flow: *The user program calls an API function. The system-call interface switches execution into kernel mode. The kernel checks arguments and permissions, performs the requested operation through a driver when needed, and returns a result or error code. Interrupts and errors can cause the kernel to handle events before returning control.*

Question 5. System-call hierarchy and related OS services



Examples and functions: *Process calls create, execute and synchronize processes; they work with scheduling. File calls open, read, write and close files, supporting storage management. Device calls request device access and configure I/O. Information calls obtain process IDs, system time and status. These calls provide the user-to-kernel boundary for OS services.*

Code of Conduct: I certify that this submission is my own work and that I have followed the assessment guidelines.

Signature: _____